Physics-Aware Multi-Objective Scheduling for Agile SAR Satellites: Balancing Profit, NESZ, and Geometric Resolution

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Abstract

In agile SAR scheduling, observation timing determines not only feasibility but also imaging quality through radar-specific geometric and radiometric effects. We characterize an empirical upper bound on multi-objective SAR quality optimization in single-pass, single-satellite Stripmap operation by integrating two SAR physical quality dimensions — Noise-Equivalent Sigma Zero (NESZ, radiometric) and geometric resolution — as decomposed optimization objectives within an NSGA-III framework, distinct from the scalar visibility composite of prior SAR-quality-aware schedulers [1] or the generic image-quality composite of Wei et al. [2]. Experiments spanning target densities from 20 to 500 reveal a scale-dependent bound: at sparse densities ($N \lesssim 100$), multi-objective optimization can extract modest geometric quality gains (+8.0% at $N = 20$) at a steep coverage cost (-31% f_1^*), while at operational densities ($N \geq 300$) profit-oriented solvers converge to a common coverage ceiling—a geometric upper bound that the ablation confirms is independent of objective-function specification. The bound is conditional on greedy-baseline hot-start initialization (material across all tested N: $\Delta f_1^* \approx -0.20$ at S1, -0.44 at S2, -0.46 at S3, -0.33 at S4, narrowing at S4), but the redundancy of the three-objective formulation (MOEA-2 $\approx$ MOEA-3) is anchor-independent. The squint-minimized greedy occupies a distinct low-coverage, high-radiometric-quality regime outside this ceiling. Rather than identifying a superior solver, our results establish an empirical upper bound on what physics-aware multi-objective scheduling can contribute to single-satellite SAR operations.

Keywords: agile SAR, satellite scheduling, multi-objective optimization, NSGA-III, noise-equivalent sigma zero (NESZ), geometric resolution

1 Introduction

Agile Earth-observation satellites equipped with Synthetic Aperture Radar (SAR) are increasingly central to time-critical monitoring missions, from disaster response to maritime surveillance. Their agility — the ability to slew the antenna beam away from the nadir track — enables a single satellite to image far more targets per orbit than a fixed-pointing platform [3]. This flexibility, however, transforms scheduling from a simple feasibility check into a combinatorial optimization problem: for N targets each with a finite visibility window, the scheduler must select a subset and assign observation times that satisfy attitude transition constraints while maximizing some measure of mission value. In optical satellite scheduling, the imaging quality of each observation is largely independent of the chosen observation time: a cloud-free nadir image of a given target has essentially the same resolution regardless of when within the visibility window it is acquired. The scheduling problem therefore reduces to maximizing the number or priority-weighted sum of observed targets subject to temporal feasibility. This assumption underpins most existing satellite scheduling work, from mixed-integer programming formulations to deep reinforcement learning approaches [4]. SAR imaging, however, violates this assumption fundamentally. SAR image quality depends on the radar geometry at the moment of acquisition: the ground-range resolution degrades with incidence angle ($\rho_{\text{rg}} \propto 1/\sin\theta$), the azimuth resolution broadens with squint angle ($\rho_{\text{az}} \propto 1/\cos\psi_{\text{sq}}$), and the radiometric sensitivity (Noise-Equivalent Sigma Zero, NESZ) deteriorates with slant range as R^3 [5]. When the scheduler can choose *when* within a visibility window to observe each target — the defining feature of agile satellites — it simultaneously chooses the imaging quality at which each target is acquired. Despite this tight coupling between scheduling and imaging physics, with few exceptions (e.g., [2, 1]), existing agile SAR scheduling methods either reduce SAR to a coverage-only objective (optimizing only task coverage) or apply post-hoc quality checks on schedules produced by coverage-maximizing solvers. Few approaches treat physical quality dimensions as primary optimization objectives in the SAR domain specifically. The problem is NP-hard by reduction from the Orienteering Problem with Time-Dependent Profits [6], precluding polynomial-time exact algorithms for all but the smallest instances. Unlike the Resource-Constrained Project Scheduling Problem (RCPSP), where task durations are fixed, our formulation features time-dependent transition times (C3) and time-dependent quality values (f_2, f_3), making the scheduling and quality-selection decisions coupled rather than separable. The SAR literature has well-established models for NESZ and geometric resolution as functions of observation geometry [3, 5], and the multi-objective optimization literature provides mature frameworks such as NSGA-III [7] for handling three or more objectives simultaneously. Yet these three bodies of knowledge — SAR physical modeling, multi-objective evolutionary algorithms (MOEAs), and satellite scheduling — have not been integrated. This paper applies MOEAs to a SAR scheduling formulation with physical quality objectives, a combination not previously reported. We formulate a three-objective agile SAR scheduling problem in which the scheduler optimizes (1) total coverage profit, (2) mean geometric image quality (ground-range and azimuth resolution), and (3) mean NESZ radiometric quality simultaneously. We compare five solvers spanning the methodological spectrum: two greedy heuristics (one profit-only baseline and one squint-minimized variant), a single-objective genetic algorithm with greedy hot-start, and two NSGA-III variants with two and three objectives. A geometric precomputation cache reduces attitude computation from $O(N^2)$ to $O(N)$ in the solver evaluation loop. The solvers are evaluated on 200 systematically varied scenarios across four density classes ($N = 20$ to $N = 500$). Our results reveal a scale-dependent bound rather than a uniform solver advantage. In sparse scenarios ($N \lesssim 100$), multi-objective optimization can extract modest geometric quality gains (+8.0% at $N = 20$) — but at a steep coverage cost (-31% f_1^*), selecting approximately 62% of the targets captured by a simple profit-only greedy scheduler. In dense scenarios

($N \geq 300$; S3–S4), the multi-objective advantage diminishes to $< 2\%$ f_2 improvement and $< 2\%$ f_1^* deficit: greedy methods approach Pareto-optimal coverage, and the squint-minimized heuristic offers competitive radiometric quality. Section 7.1 provides a detailed synthesis of these findings. The specific contributions of this work are:

1. The decomposition of SAR image quality into closed-form NESZ ($\propto R^3$) and geometric resolution ($\sin\theta \cos\psi_{\text{sq}}$) objectives within an NSGA-III framework—distinct from the scalar visibility composite of Kramer & Miller [1] and the generic image-quality composite of Wei et al. [2]—using mean per-task quality metrics that decouple quality from the number of scheduled tasks;
2. The discovery and analytical characterization of an empirical upper bound on multi-objective SAR quality optimization in the single-pass, single-satellite Stripmap setting: the bound arises from solver convergence to low-squint regimes (where the empirical $r = 0.93$–0.98 exceeds the definitional null $r_{\text{null}} = -0.51$, § 3.2), not from a physical identity between objectives, and renders the three-objective formulation redundant at operational densities ($N \geq 300$). A scale phase transition concentrates the multi-objective advantage in sparse regimes (+8.0% geometric quality at $N = 20$), beyond which profit-oriented solvers converge to a common coverage ceiling;
3. A systematic empirical characterization using five solvers (greedy, single-objective GA, and NSGA-III variants) across four scale classes and an ablation study covering four objective-function variants, confirming that physical quality objectives affect solver outcomes only in sparse regimes and that the bound is conditional on greedy-baseline hot-start initialization.

2 Related Work

We review three lines of research that our work bridges: satellite scheduling (2.1), SAR physical quality modeling (2.2), and multi-objective evolutionary optimization (2.3).

2.1 Satellite Scheduling

Satellite scheduling has been studied extensively in the Earth-observation domain. The survey by Ferrari et al. [8] catalogs over 200 publications spanning exact methods, heuristics, metaheuristics, and learning-based approaches. Exact formulations based on mixed-integer linear programming (MILP) achieve optimality on small instances but fail to scale to operational problem sizes, motivating a broad family of heuristic and metaheuristic alternatives including genetic algorithms, simulated annealing, and tabu search [9, 10]. The earliest formal treatments of agile satellite scheduling by Lemaître et al. [11] and Bianchessi et al. [12] established the foundational problem structure of time-dependent transition times between observations. Recent work further generalizes these methods to agile satellites, where the freedom to choose observation start times within each visibility window is exploited to pack more observations into a single orbit [6, 13]. A parallel line of research has pursued learning-based and hybrid approaches for satellite scheduling. Liu et al. [14] and Zhou et al. [15] apply deep reinforcement learning to agile optical satellite scheduling, while Jacquet et al. [16] combine graph neural networks with Monte Carlo tree search for the same domain. Wang et al. [17] propose a Lagrangian relaxation-based heuristic for multi-satellite mission planning. Caleiras et al. [18] formulate super-agile satellite scheduling as a constraint programming problem, capturing the time-dependent transition constraints characteristic of super-agile platforms. Herrmann et al. [19] apply RL to on-board AEOS scheduling, maximizing target collection reward without imaging quality as an objective. Wu et al. [20] address high-target-density AEOS scheduling with an improved whale optimization algorithm, reporting convergence challenges that echo the density-driven constraint tightening we analyze in § 6.5. Despite their methodological diversity, all these works treat quality as a fixed property determined at observation-window generation. In contrast, SAR imaging quality varies continuously with acquisition geometry and is directly optimizable within the scheduling window. Chang et al. [21] propose the first MOEA for SAR satellite scheduling using two adaptive NSGA-II variants on a Gaofen-12 X-band SAR platform; their bi-objective formulation (profit + energy consumption) represents the closest prior work to ours, but it omits imaging quality dimensions. Wei et al. [2] introduce a neural-policy-guided MOEA that incorporates a composite image quality score as an optimization objective. Kramer & Miller [1] schedule SAR constellation observations with a hybrid metaheuristic using a weighted visibility–performance composite, embedding a SAR-quality proxy in the scheduler, and Yao et al. [22] develop a bilevel evolutionary algorithm for multi-satellite scheduling that separates task allocation from detailed observation planning. The common limitation of all this prior work is that SAR physical quality metrics — noise-equivalent sigma zero (NESZ), azimuth resolution, ground-range resolution — are either absent or approximated by coarse proxies (e.g., penalizing extreme off-nadir angles). No existing SAR scheduling method treats continuous, physics-derived quality functions as optimization objectives in their own right.

2.2 SAR Physical Quality Modeling

The SAR remote sensing literature provides well-established analytical models linking observation geometry to image quality. Curlander and McDonough [3], Cumming and Wong [5], and Moreira et al. [23] give comprehensive treatments of SAR physics, including the two quality dimensions relevant to scheduling: geometric resolution and radiometric sensitivity. Geometric resolution decomposes into ground-range resolution $\rho_{\mathrm{rg}} = c/(2B\sin\theta)$, where c is the speed of light, B the radar bandwidth, and θ the incidence angle, and azimuth resolution $\rho_{\mathrm{az}} = D/(2\cos\psi_{\mathrm{sq}})$, where D is the antenna length and ψ_{sq} the squint angle. This gives the unweighted theoretical limit; in operational SAR, Taylor or Hamming amplitude weighting widens ρ_{az} by 10–20% [3]. Since this weighting is platform-constant, the unweighted form preserves the relative ordering of observation geometries for optimization purposes. Both degrade as the observation departs from the zero-Doppler geometry ($\psi_{\mathrm{sq}} = 0$, θ minimal). Radiometric sensitivity, measured by the Noise-Equivalent Sigma Zero (NESZ), deteriorates with the cube of slant range: $\mathrm{NESZ} \propto R^3 \propto 1/(\cos^3\theta \cdot \cos^3\psi_{\mathrm{sq}})$ for Stripmap SAR after azimuth compression gain [3]. The proportionality $\mathrm{NESZ} \propto R^3$ captures the dominant geometric dependence; noise figure, transmit power, and processing gain terms are omitted as they are platform-constant and do not affect the relative ordering of observation geometries that the scheduler optimizes over. This cubic dependence on slant range means even modest deviations from the optimal viewing geometry make fine observation timing critical for radiometric performance. Moreira et al. [23] survey the evolution of spaceborne SAR technologies and reinforce that imaging quality is intrinsically tied to acquisition geometry. These quality models are routinely used in SAR system design to specify performance envelopes (e.g., maximum NESZ across a swath). Their use in scheduling, however, remains limited to post-hoc validation of fixed observation plans [24] or binary feasibility filtering at the observation-window level [25] (arXiv preprint). The scheduling literature and the SAR quality modeling literature have developed in parallel: schedulers treat quality as a constraint, while SAR engineers compute quality after the schedule is fixed. Our work connects these two domains by embedding continuous quality functions directly into the optimization objective space.

2.3 Multi-Objective Evolutionary Optimization

Multi-objective evolutionary algorithms (MOEAs) have matured into a standard toolkit for problems with two or more competing objectives. The NSGA-III algorithm [7] extends the well-known NSGA-II framework to many-objective problems ($M \geq 3$) through a reference-point-based diversity preservation mechanism, replacing the crowding distance operator that degrades in higher dimensions. NSGA-III has been applied to aerospace trajectory optimization, constellation design, and mission planning problems where objectives such as fuel consumption, coverage, and revisit time must be balanced simultaneously. In satellite scheduling, MOEAs have been applied primarily in optical domains, with objectives that include total profit, task completion time, resource utilization, and load balancing [26, 27]. Yang et al. [28] survey recent developments in surrogate-assisted expensive MOEAs for multi-satellite imaging mission planning. The hypervolume (HV) indicator is the most widely used quality metric for comparing Pareto front approximations, but its sensitivity to frontier cardinality creates interpretational challenges when comparing single-objective and multi-objective solvers — a methodological issue we address in § 6.3. The decompose-and-learn framework of Qi et al. [29] tackles large-scale satellite scheduling with a multi-objective decomposition strategy, demonstrating that MOEA scalability can be achieved through problem-specific decomposition. To our knowledge, MOEAs have not been applied to SAR scheduling with physical quality objectives. The closest work uses single-objective optimization with post-hoc quality evaluation, which cannot reveal the structure of trade-offs between coverage and imaging quality. Our NSGA-III formulation makes these trade-offs explicit by placing coverage profit, geometric quality, and radiometric quality in a unified multi-objective framework, producing Pareto fronts that mission planners can inspect to make informed quality–coverage decisions.

3 Problem Formulation

3.1 System Model

We consider a single agile SAR satellite in a circular orbit at altitude H, operating in Stripmap mode. We adopt Stripmap as the baseline SAR mode because it provides the widest continuous swath with uniform quality, and its quality–geometry relationships are analytically tractable through closed-form NESZ and resolution expressions [3]. Extension to ScanSAR and TOPS modes is discussed in § 7.4. A problem instance $\mathcal{I} = (\mathcal{T}, \mathcal{S}, \mathcal{O})$ consists of a set of N ground targets $\mathcal{T}$, a satellite platform $\mathcal{S}$, and a SAR imaging model $\mathcal{O}$.

Task Set. Each task $i \in \{1, \ldots, N\}$ is specified by its geographic coordinates $\mathbf{r}_i = (\phi_i, \lambda_i)$, a priority weight $p_i \in \mathbb{R}^+$, a minimum required ground resolution ρ_i^{req}, a visibility window $\mathcal{W}_i = [a_i, b_i]$ during which the satellite can observe the target, and a fixed observation duration d_i.

Satellite Platform. The satellite is characterized by its maximum slew rate $\omega_{\max}$ (a conservative single-axis bound on three-axis attitude maneuvers), settling time τ_{settle} after each maneuver, per-orbit energy budget $E_{\max}$ and memory budget $M_{\max}$, and per-observation energy e_i and memory m_i consumption.

SAR Observables. Given an observation time t_i, the satellite-to-target line-of-sight (LOS) vector $\mathbf{l}(t_i)$ in the Local-Vertical Local-Horizontal (LVLH) frame is fully determined by the satellite's orbital state. From $\mathbf{l}(t_i)$ we derive four geometric quantities deterministically: slant range $R = |\mathbf{l}|$, off-nadir angle $\phi = \arctan 2(\sqrt{l_x^2 + l_y^2},\, l_z)$, incidence angle θ (via the Earth-curvature-corrected mapping $\theta = \arcsin\left(\frac{R_e+H}{R_e}\sin\phi\right)$ where R_e is Earth's mean radius), and squint angle $\psi_{\text{sq}} = \arcsin(|l_x|/R)$. This definition of squint angle as the angular deviation of the LOS vector from the zero-Doppler plane follows Curlander and McDonough [3] (Ch. 9, SAR Doppler characteristics), where the azimuth Doppler centroid is proportional to $\sin\psi_{\text{sq}}$. A critical property is that *all* quality-relevant geometry is a deterministic function of the single decision t_i.

Decision Variables. A schedule consists of a task selection vector $\mathbf{x} \in \{0,1\}^N$, an ordered sequence $S = (s_1, \ldots, s_m)$ of the m selected tasks, and a vector of observation start times $\mathbf{t} = (t_{s_1}, \ldots, t_{s_m})$. All geometric parameters (θ_i, $\psi_{\text{sq},i}$, R_i) are *derived* from t_i rather than independent decision variables.

3.2 Objective Functions

We formulate a three-objective optimization problem that captures the tension between scheduling throughput and SAR-specific imaging quality. All quality objectives use the *mean* over scheduled tasks, making them independent of the number of scheduled tasks $m = |\mathcal{S}|$ and thereby producing a genuine "quantity versus quality" Pareto trade-off.

Objective 1: Coverage Profit. The normalized coverage profit (Eq. 1) is defined relative to the greedy baseline

$$f_1^* = \frac{1}{f_1^{\text{G-BL}}} \sum_{i \in \mathcal{S}} p_i \tag{1}$$

where $f_1^{\text{G-BL}}$ is the profit achieved by the profit-only greedy scheduler (G-BL, § 4.2) on the same scenario.

Objective 2: Geometric Image Quality. The ground pixel area of a SAR image is the product of ground-range resolution $\rho_{\text{rg}} = c/(2B\sin\theta)$ and azimuth resolution $\rho_{\text{az}} = D/(2\cos\psi_{\text{sq}})$. We define a dimensionless geometric quality index:

$$f_2 = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \sin\theta_i \cdot \cos\psi_{\text{sq},i} \tag{2}$$

Higher f_2 indicates smaller ground pixel area (better geometric resolution). When $\mathcal{S} = \emptyset$, $f_2 = 0$.

Objective 3: NESZ Radiometric Quality. Noise-Equivalent Sigma Zero (NESZ) measures the minimum detectable backscatter. For Stripmap SAR, NESZ degrades with the cube of slant range, NESZ $\propto R^3 \propto 1/(\cos^3\theta \cdot \cos^3\psi_{\text{sq}})$, after accounting for azimuth compression gain [3]. We define the radiometric quality index as:

$$f_3 = \frac{1}{|\mathcal{S}|}\sum_{i\in\mathcal{S}} \cos^3\theta_i \cdot \cos^3\psi_{\text{sq},i} \tag{3}$$

Higher f_3 indicates lower NESZ (better radiometric sensitivity). When $\mathcal{S} = \emptyset$, $f_3 = 0$. Both f_2 and f_3 are bounded to $[0,1]$ by construction: $\sin\theta_i$, $\cos\psi_{\text{sq},i}$, and $\cos\theta_i$ all lie within $[0,1]$ for the attitude envelope used in this study.

Physical Trade-Off. The two quality objectives partition SAR imaging performance by physical mechanism: f_2 captures spatial resolution (geometric), f_3 captures signal-to-noise ratio (radiometric). Their optima are not coincident — f_3 peaks at the zero-Doppler point (θ minimal, $\psi_{\text{sq}} = 0$), whereas f_2 benefits from moderately higher θ because $\sin\theta$ grows with incidence angle. This creates a genuine trade-off: pursuing the cleanest NESZ (zero-Doppler observation) forces acceptance of stretched ground-range pixels; pursuing optimal ground pixel area requires deviating from the zero-Doppler point and accepting elevated radiometric noise. Critically, the squint angle ψ_{sq} appears in both f_2 and f_3 (Eqs. 2–3): reducing ψ_{sq} simultaneously improves both resolution components, but at the expense of f_2 through reduced $\sin\theta$ — creating the $[\theta, \psi_{\text{sq}}]$ geometric structure that bounds multi-objective gain once solvers converge to low-squint regimes (see Eq. 4 below and its axis decomposition for the exact partition into definitional and scheduler-induced components).

Because θ and ψ_{sq} are not independent but both derived from the same LOS vector $\mathbf{l} = (l_x, l_y, l_z)$ (§ 3.1), the per-task f_2 and f_3 contributions share geometric structure. To separate definitional coupling from scheduler-induced correlation, we compute the analytical null distribution by treating observation geometry as drawn uniformly from the accessible envelope ($\theta \in [15^\circ, 50^\circ]$, $|\psi_{\text{sq}}| \le 45^\circ$) and evaluating the moment integrals of $f_{2,i} = \sin\theta_i \cos\psi_{\text{sq},i}$ and $f_{3,i} = \cos^3\theta_i \cos^3\psi_{\text{sq},i}$ in closed form (validated by Monte Carlo, $r_{\text{MC}} = -0.5120$, $N = 2\times10^5$). The null correlation is

$$r_{\text{null}} = \frac{\operatorname{Cov}(f_2, f_3)}{\sqrt{\operatorname{Var}(f_2)\operatorname{Var}(f_3)}} \;=\; -0.51, \tag{4}$$

where the independence of θ and ψ_{sq} under the null factorizes the moments, e.g., $E[f_2 f_3] = E[\sin\theta\cos^3\theta]\,E[\cos^4\psi_{\text{sq}}]$. The *negative* sign exposes a structural conflict, not redundancy: in the θ-only limit (fixed squint), $f_{2,i} \propto \sin\theta$ and $f_{3,i} \propto \cos^3\theta$ are strictly anti-monotonic ($r_\theta = -0.997$), since f_2 favors high incidence while f_3 favors low incidence; in the ψ_{sq}-only limit (fixed θ), both depend on $\cos\psi_{\text{sq}}$ positively ($r_\psi = +0.997$), since low squint improves both. Under uniform geometry the θ-conflict dominates, yielding $r_{\text{null}} < 0$. The empirical correlation $r \approx 0.93$–0.98 (§ 6.3) therefore exceeds the definitional baseline substantially: in Fisher-z space, $z(r_{\text{null}} = -0.51) = -0.563$ while $z(r) = 1.66$–2.30 for the empirical range, a gap of $\Delta z \approx 2.2$–2.9 that confirms the empirical correlation lies far above the null (correlation distances are compared on the variance-stabilizing Fisher-z scale rather than as a linear difference). This indicates that scheduled observations are not drawn uniformly from the envelope: all solvers, including the greedy baselines, select low-squint geometries that suppress the θ-conflict and align f_2 with f_3 along the shared $\cos\psi_{\text{sq}}$ factor.

The residual 2–7% variance $(1-r)$ corresponds to the θ-axis conflict space where genuine f_2–f_3 trade-offs can exist, but which the solver-converged low-squint regime leaves largely unexplored in the single-pass setting. This reframes the 3-objective redundancy reported in § 6.3: it is not a mathematical identity between f_2 and f_3 but an empirical consequence of solver convergence to low-squint regimes.

The visible-geometry envelope is wide rather than low-squint-dominated: across C7-feasible visibility samples ($|\psi_{\text{sq}}| \leq 45°$), $\text{corr}(f_2, f_3) = -0.95$, a strong negative correlation dominated by the incidence-angle conflict (`scripts/r_visible_envelope.py`). In contrast, solver-selected schedules exhibit $r = +0.93$–0.98, indicating that the low-squint regime of solver schedules is an active selection from the wide C7-feasible envelope, not a constraint-imposed collapse. This supports the solver-convergence interpretation: all tested solvers (including the no-physics Variant D, § 6.7) converge to low-squint regimes where f_2 and f_3 co-vary along the shared $\cos\psi_{\text{sq}}$ factor, rendering the 3-objective formulation redundant in the single-pass setting.

3.3 Constraints

Seven constraints couple task selection with temporal feasibility and resource limits. Constraints C1, C2, and C7 are enforced during visibility window generation; C3–C6 are checked inline by every solver.

C1: Visibility Window. Equation (5) defines the visibility-window constraint. Each observation must fit within a pre-computed visibility window:

$$t_i \in [a_i, b_i - d_i] \quad \forall i \in \mathcal{S} \tag{5}$$

C2: Resolution Requirement. The achieved ground-range resolution must meet the task's requirement. Via $\rho_{\text{rg}} = c/(2B\sin\theta)$, this imposes a lower bound on incidence angle: $\theta_i \geq \theta_i^{\min}$ where $\theta_i^{\min} = \arcsin\left(c/(2B\rho_i^{\text{req}})\right)$.

C3: Attitude Maneuver Transition. Equation (6) defines the attitude-maneuver transition constraint. Between consecutive observations, the satellite must rotate through the angular separation of their LOS vectors. Let $\mathbf{l}_k$ and $\mathbf{l}_{k+1}$ be the LVLH-frame LOS vectors at t_{s_k} and $t_{s_{k+1}}$. The angular gap $\Delta\eta_k = \arccos(\mathbf{l}_k \cdot \mathbf{l}_{k+1}/(|\mathbf{l}_k||\mathbf{l}_{k+1}|))$ must be traversed within the available inter-observation time:

$$t_{s_{k+1}} \geq t_{s_k} + d_{s_k} + \frac{\Delta\eta_k}{\omega_{\max}} + \tau_{\text{settle}} \quad \forall k \in \{1, \ldots, m-1\} \tag{6}$$

This constraint is the *core coupling* in our formulation: different choices of t_{s_k} shift the LOS vector $\mathbf{l}_k$, affecting both imaging quality (via θ_k, $\psi_{\text{sq},k}$) and transition feasibility to subsequent tasks (via $\Delta\eta_k$).

C4: Energy Budget. $\sum_{i\in\mathcal{S}} e_i \leq E_{\max}$, where $E_{\max}$ is the aggregate energy budget for the orbit.

C5: Memory Budget. $\sum_{i\in\mathcal{S}} m_i \le M_{\max}$, where $M_{\max}$ is the aggregate on-board storage budget. The values $E_{\max} = 10^7$ and $M_{\max} = 10^{11}$ (arbitrary units) are used in all experiments; these budgets are set sufficiently high to be non-binding under all tested scenarios, i.e., they never eliminated a solution candidate in any solver run. They are retained in the formulation for completeness, so that the framework can accommodate operational energy and storage budgets when they become available.

C6: No Overlap. Implied by C3; no two observations may overlap in time.

C7: Squint Angle Limit. $|\psi_{\mathrm{sq},i}| \le \psi_{\mathrm{sq}}^{\max}$ for all $i \in \mathcal{S}$, where $\psi_{\mathrm{sq}}^{\max}$ is the satellite's maximum achievable squint (typically $\pm 45°$). The tightness of C3 scales with task density: as N grows, the fraction of task pairs whose required angular transition $\Delta\eta/\omega_{\max}$ exceeds the available inter-observation time increases, tightening the constraint. At S1 ($N = 20$), approximately $< 10\%$ of task pairs violate C3 under the maximum maneuver rate; at S4 ($N = 500$), this fraction rises to 30–40%, effectively making C3 the dominant active constraint in dense regimes.

3.4 Geometric Precomputation

Evaluating attitude-dependent constraints (C3) and quality objectives (f_2, f_3) during optimization would require on-the-fly orbit propagation and LVLH-frame coordinate transforms at every solver evaluation step, imposing a $O(N^2 \cdot K)$ overhead for N tasks and K discretized time samples per window. We eliminate this bottleneck through a *geometric precomputation cache* (GeomCache) that computes, for every task i at every feasible discretized observation time within its window, the derived geometric quantities θ_i, $\psi_{\mathrm{sq},i}$, R_i, ϕ_i, and the inter-task LOS angular gaps $\Delta\eta_{ij}$. Following this one-time $O(N^2 \cdot K)$ precomputation, all solver operations access geometry via $O(1)$ table lookups. During MOEA evaluation, attitude feasibility and quality scores reduce from $O(N^2)$ to $O(N)$ per individual through the precomputed geometry cache, reducing the asymptotic complexity of the innermost optimization loop.

4 Methodology

4.1 Solver Overview

Table 1 summarizes the five solvers compared in this study. We present the solvers in order of increasing sophistication: from greedy heuristics (G-BL, G-SM) through single-objective metaheuristic (GA-P-BL), to multi-objective evolutionary algorithms (MOEA-2, MOEA-3).

Table 1: Solvers used in this study

Label	Type	Objectives	Hot-Start
G-BL	Greedy heuristic	max f_1 (profit-only)	—
G-SM	Greedy heuristic	max f_1 + squint minimization	—
GA-P-BL	Single-objective GA	max f_1 (G-BL seeded)	std = 0.5
MOEA-2	NSGA-III (2-objective)	$f_1^* + f_2$ (geometric quality)	std = 0.5
MOEA-3	NSGA-III (3-objective)	$f_1^* + f_2 + f_3$ (geo.+NESZ)	std = 0.5

All five solvers share the geometric precomputation cache (§ 3.4) for $O(1)$ access to θ_i, $\psi_{\mathrm{sq},i}$, and $\Delta\eta_{ij}$. Single-objective solvers (G-BL, G-SM, GA-P-BL) optimize only f_1; their f_2 and f_3 values are computed post-hoc on the final schedule for comparison with the multi-objective solvers. We note that the five solvers differ primarily in objective-function configuration rather than algorithmic paradigm. This design is intentional: it isolates the effect of adding physical quality objectives, which is the paper's research question. A comparison across algorithmic families (e.g., NSGA-III vs. MOEA/D vs. SMS-EMOA) is left to future work (§ 7.4).

4.2 G-BL: Greedy Baseline

G-BL represents the conventional optical-domain scheduling approach: it makes no use of SAR imaging physics when ordering or selecting tasks. Tasks are sorted by earliest start time t_{start} and greedily inserted into the schedule at their earliest feasible time. Each task is observed at the off-nadir angle pre-stored in its visibility window (the angle yielding maximum elevation — equivalent to a fixed-nadir policy). If the attitude transition constraint (C3) is violated for a candidate task, its start time is delayed within the window until the constraint is satisfied; if no feasible delay exists, the task is dropped. Since G-BL selects observation times solely for feasibility, its f_2 and f_3 values are incidental by-products of the window geometry. We compute these post-hoc on the final schedule to serve as the baseline against which physics-aware solvers are compared.

4.3 G-SM: Greedy Squint-Minimized

G-SM shares G-BL's greedy structure (sort by earliest start time) but modifies the observation time selection: within each visibility window, a task is placed at the discretized time step that minimizes squint angle $\psi_{\mathrm{sq}} = \arcsin(|l_x|/R)$. G-SM minimizes squint angle as a computationally cheap proxy for maximizing f_3 (NESZ radiometric quality); the correlation between squint minimization and f_3 improvement is confirmed by the large effect size (Cliff's $\delta = +0.72$, § 7.1). When the window contains the zero-Doppler crossing ($l_x = 0$), G-SM achieves $\psi_{\mathrm{sq}} = 0$ and the theoretical azimuth resolution limit $\rho_{\mathrm{az}} = D/2$. With 60-s orbit propagation resolution, the true zero-Doppler instant may fall between samples; G-SM selects the discretized time that minimizes $|\psi_{\mathrm{sq}}|$, which may yield $|\psi_{\mathrm{sq}}| \leq 0.5°$ rather than exactly 0° for windows that do not align with a sample point. This strategy simulates the theoretical Stripmap SAR operating mode and serves as a physics-aware greedy baseline: it answers the question "what quality improvement can a simple squint-minimization heuristic achieve, without any multi-objective search?" The squint-minimizing time selection incurs a profit penalty because the optimal-squint time may lie later in the window, reducing the remaining time for subsequent tasks (constraint C3). As with G-BL, f_2 and f_3 are post-hoc outputs.

4.4 GA-P-BL: Single-Objective GA with Hot-Start

GA-P-BL tests whether a genetic algorithm optimizing only f_1 can outperform greedy scheduling by exploiting temporal flexibility. Each chromosome encodes a vector of normalized start times $\boldsymbol{\tau} \in [0,1]^N$, where task i's actual start time is $t_i = t_i^{\text{earliest}} + \tau_i \cdot (t_i^{\text{latest}} - d_i - t_i^{\text{earliest}})$. The task selection vector $\mathbf{x}$ emerges from constraint resolution: a task is dropped if no feasible insertion time exists within its window after satisfying C3. The population of 100 individuals is initialized from G-BL's solution with Gaussian noise of standard deviation 0.5 added to each τ_i. This noise level is calibrated to allow task selections to change — smaller noise (e.g., 0.02) locked the task set to G-BL's selection, preventing the GA from discovering profit-improving task substitutions. The GA runs for 200 generations with simulated binary crossover ($p_c = 0.9$, $\eta_c = 20$) and polynomial mutation ($p_m = 0.1$, $\eta_m = 20$). f_2 and f_3 are computed post-hoc on the best-found schedule.

4.5 MOEA-2 and MOEA-3: Multi-Objective Evolutionary Algorithms

MOEA-2 and MOEA-3 extend the GA-P-BL chromosome encoding to multi-objective optimization via the NSGA-III framework [7]. Both use a population of 100 individuals over 200 generations with the same genetic operators as GA-P-BL. Reference points are generated via the Das–Dennis systematic approach with 12 divisions per objective dimension, producing $H = \binom{M+p-1}{p}$ structured reference directions ($H = 13$ for MOEA-2, $H = 91$ for MOEA-3). The entire population is hot-started from the G-BL solution perturbed by Gaussian noise (std $= 0.5$), and an at-least-one-task constraint prevents empty solutions from dominating the Pareto front. MOEA-2 optimizes two objectives: f_1^* (normalized profit) and f_2 (mean geometric quality, Eq. 2). MOEA-3 adds f_3 (mean NESZ radiometric quality, Eq. 3) as a third objective. The comparison between MOEA-2 and MOEA-3 directly tests whether adding the NESZ objective (f_3) produces Pareto-front differentiation beyond what f_2 alone captures, or whether the θ/ψ_{sq} geometric coupling renders the third objective redundant in the single-pass, single-satellite setting.

5 Experimental Setup

5.1 Scenario Design

Table 2 defines the four scenario classes.

Table 2: Scenario classes

Class	Targets	Look Direction	Description
S1	20	Both	Small, low-density
S2	100	Both	Medium, moderate density
S3	300	Both	Medium-large, transition region
S4	500	Both	Large, high-density

All scenarios share a common satellite platform abstracted as a generic C-band Stripmap SAR with bilateral-looking capability: a circular sun-synchronous orbit at 600 km altitude (inclination $\approx 97.8°$, LTAN 06:00, dawn–dusk), operating in Stripmap mode with a C-band SAR instrument (antenna length $D = 12.3$ m, bandwidth $B = 56.5$ MHz). The platform accesses both left- and right-side incidence angles in the range $[18°, 47°]$ (signed convention: $\theta \in [-47°, -18°] \cup [18°, 47°]$). We note that this bilateral Stripmap abstraction departs from operational Sentinel-1A/B, which operates primarily in right-looking TOPS mode (Interferometric Wide swath, incidence $\sim 29°$–$46°$) [30]; the bilateral-looking Stripmap configuration here represents a generic agile-SAR concept (cf. TerraSAR-X dual-receive Stripmap [31] and prospective bistatic formations) selected to exercise the full incidence/squint trade-off space. Targets are distributed along the satellite ground track within the accessible swath (cross-track scatter $\pm 2.7°$, roughly ± 300 km), with three distribution types per scenario class: uniform random, clustered (5 clusters), and mixed (half uniform, half clustered). The five scenarios within each class correspond to distribution type and random-seed variations; the 50 scenarios per class therefore consist of 5 distribution–seed combinations $\times$ 10 random seeds. The four density classes (S1–S4) span the operational range from sparse reconnaissance ($N = 20$) to dense monitoring ($N = 500$); intermediate densities ($N = 50, 150, 200, 400$) are omitted to limit computational cost, with the transition region $N = 100$–300 analyzed through the scale-sensitivity experiments in § 6.5. Visibility windows are computed via orbit propagation at 60-s resolution (producing $K \approx 200$ discrete time samples per orbit) over two orbital periods ($\sim$200 min), yielding 1–2 windows per target on average, each spanning several minutes (typically 3–8 samples per window). Windows are filtered by incidence-angle and minimum-elevation (5°) constraints; the resulting window set encodes the bilateral geometry that the solver must navigate: targets are roughly evenly split between left- and right-looking windows ($\sim$50% each), and cross-side attitude transitions (e.g., $+30° \to -30°$) incur substantially larger angular gaps than same-side transitions. Approximately 50%–60% of all visibility windows contain a zero-Doppler crossing ($\psi_{\mathrm{sq}} = 0$), providing G-SM with at least one sample point per window where the theoretical azimuth resolution limit is attainable; the fraction varies with target latitude and orbital position but remains above 40% across all scenario classes.

5.2 Solver Configuration

All five solvers use the geometric precomputation cache (§ 3.4) and run on a single CPU core to ensure reproducible timing comparisons. G-BL and G-SM are deterministic greedy heuristics with no tunable hyperparameters. GA-P-BL uses a population of 100 individuals over 200 generations with simulated binary crossover ($p_c = 0.9$, $\eta_c = 20$) and polynomial mutation ($p_m = 0.1$, $\eta_m = 20$), hot-started from the G-BL solution as described in § 4.4. MOEA-2 and MOEA-3 employ NSGA-III with the same population size, generation budget, and genetic operators as GA-P-BL. Reference directions are generated via Das–Dennis systematic sampling with 12 partitions per objective dimension ($H = 13$ for two objectives, $H = 91$ for three). With a population of 100 individuals and $H = 91$ reference directions for MOEA-3, approximately one individual is assigned per direction; niching pressure may therefore be weak. We verified that increasing the population to 200 does not qualitatively change the results reported below. The entire MOEA population is hot-started from the G-BL solution perturbed by Gaussian noise ($\sigma = 0.5$), and an at-least-one-task constraint prevents the empty schedule from dominating the Pareto front.

5.3 Evaluation Metrics

We use hypervolume (HV) as the primary quality indicator for multi-objective comparison. Statistical significance is assessed via:

- **Friedman test**: overall difference across 5 solvers $\times$ 200 scenarios
- **Wilcoxon signed-rank**: pairwise post-hoc comparisons with Bonferroni correction. The primary family comprises 10 pairwise HV comparisons ($\alpha = 0.05/10 = 0.005$); per-objective pairwise tests (6 additional) and ablation comparisons (3 variants $\times$ 4 classes $\times$ 6 metrics) are reported with raw p-values alongside the Bonferroni threshold, so readers can evaluate significance under alternative corrections (e.g., Benjamini–Hochberg FDR) or tighter family-wise correction.
- **Cliff's delta**: non-parametric effect size, classified by the Vargha–Delaney convention ($|\delta| < 0.147$ negligible, < 0.33 small, < 0.474 medium, ≥ 0.474 large). As operational anchors, a $|\delta| \geq 0.474$ (large) on f_1^* corresponds to a $\geq 15\%$ coverage difference at S4 (~ 30 tasks); a $|\delta| < 0.147$ (negligible) on f_2 corresponds to < 0.01 Δf_2, equivalent to < 0.5 m ground-range resolution at C-band.

Hypervolume is computed with reference point $(0, 0, 0)$ in the per-solver-pooled min–max-normalized objective space (all solvers' f_i values and frontier points are pooled to define a shared $[0, 1]$ scale; see `experiments/statistical_analysis.py`).

6 Results and Analysis

6.1 Effect of Squint Awareness

The squint-minimized greedy solver (G-SM) improves NESZ radiometric quality (f_3) relative to the profit-only baseline (G-BL) by selecting observation times that minimize squint angle within each visibility window. Across all 200 scenarios, this produces a large positive effect on f_3 (Cliff's $\delta = +0.72$, $p = 1.7 \times 10^{-34}$) at a small aggregate cost in coverage (Cliff's $\delta = -0.25$, $p = 7.0 \times 10^{-34}$, on raw f_1). The trade-off, however, is strongly density-dependent. In sparse scenarios (S1, $N = 20$), G-SM retains $f_1^* = 0.85 \pm 0.16$ (-15% vs. G-BL at 1.00) while improving f_3 from 0.32 ± 0.17 to 0.44 ± 0.16 (+34%). In dense scenarios (S4, $N = 500$), the f_1 cost escalates: G-SM achieves $f_1^* = 0.61 \pm 0.09$ (-39%), with f_3 rising from 0.210 ± 0.071 to 0.356 ± 0.075 (+70%). Figure 1 visualizes this density-dependent trade-off: the LOESS trend in panel (c) identifies a transition region around $N = 100$ where the f_1 penalty grows rapidly. In dense regimes, the squint-minimization strategy becomes costly because tightly packed time windows offer fewer alternative observation times at low squint.

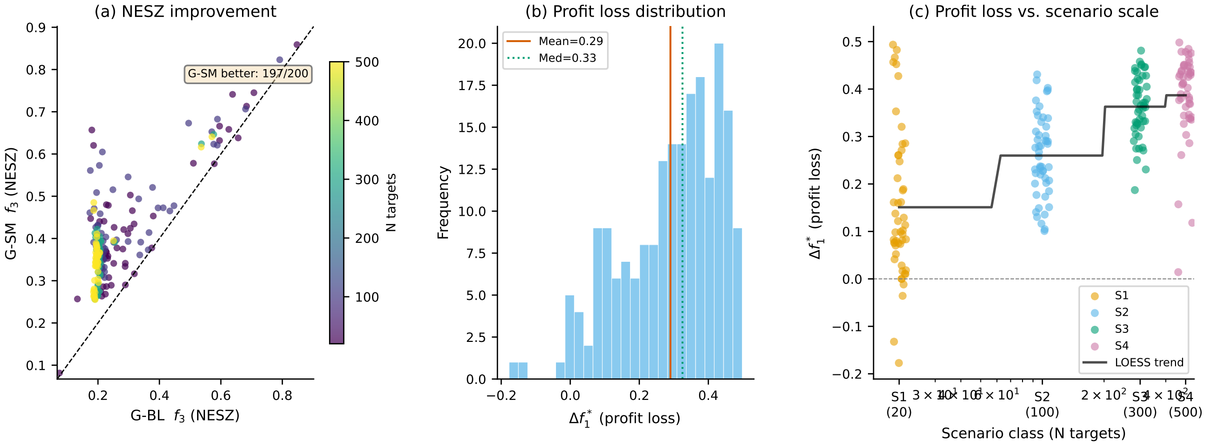

Figure 1: Effect of squint-aware scheduling: (a) f_3 scatter; (b) Δf_1^* histogram; (c) Δf_1^* vs. N with LOESS trend

6.2 Solver Profiles

Table 3 summarizes the three-dimensional performance of all five solvers on the largest scenario class (S4, $N = 500$), where differences are most compressed and therefore hardest to achieve. G-BL achieves the baseline $f_1^* = 1.00$ with $f_2 = 0.580 \pm 0.025$ and $f_3 = 0.210 \pm 0.071$. G-SM trades f_1 for f_3 quality: $f_1^* = 0.61 \pm 0.09$, $f_3 = 0.356 \pm 0.075$, with $f_2 = 0.506 \pm 0.035$ — notably lower than all other solvers on geometric quality. GA-P-BL surpasses the greedy baseline on f_1 ($f_1^* = 1.17 \pm 0.27$) with numerically similar f_2 and f_3 (0.580 ± 0.024 and 0.203 ± 0.073, respectively), confirming that the single-objective GA exploits time flexibility that the greedy scheduler misses. GA-P-BL achieves $f_1^* > 1.0$ in 58% (S1: 29/50) to 100% (S2: 50/50) of scenarios, with mean f_1^* ranging from 1.09 ± 0.14 at S1 to 1.17 ± 0.27 at S4. Crucially, GA-P-BL maintains f_2 and f_3 quality comparable to G-BL: $\Delta f_2 \in [-0.001, +0.009]$ and $\Delta f_3 \in [-0.030, -0.007]$ across all scales. The coverage improvement therefore does not come at a quality cost — GA-P-BL genuinely identifies schedules with higher task counts while preserving per-task geometric and radiometric quality, challenging the claim that greedy scheduling is Pareto-sufficient. MOEA-2 and MOEA-3 both achieve near-baseline coverage ($f_1^* = 0.98 \pm 0.03$ and 0.98 ± 0.04) while improving f_2 to 0.589 ± 0.021. f_3 drops to 0.169 ± 0.066 and 0.170 ± 0.066, respectively (vs. G-BL 0.210 ± 0.071; Cliff's $\delta = -0.63$, large negative effect for MOEA-2). The near-identity of MOEA-2 and MOEA-3 results is analyzed in detail in § 6.3. Figure 2 compares MOEA-2 and MOEA-3 Pareto fronts on an S4 scenario, confirming that the solver ranking is scale-dependent: GA-P-BL maximizes f_1^* at no quality cost, while the MOEA solvers occupy a balanced middle ground on all three dimensions.

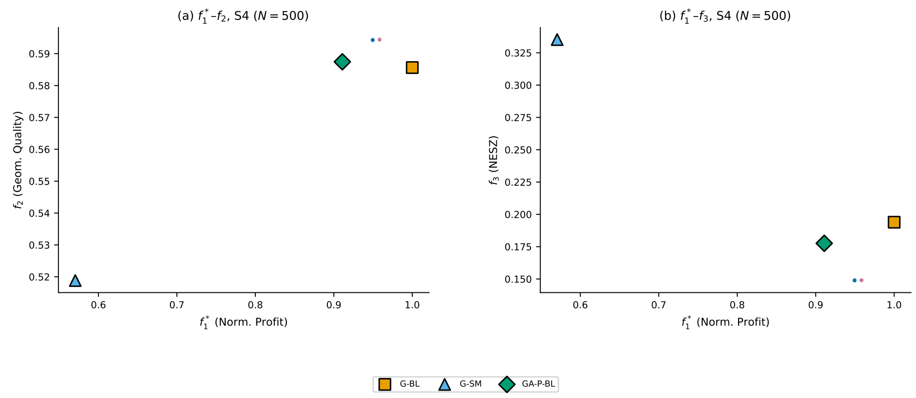

Figure 2: MOEA-2 vs. MOEA-3 Pareto fronts on a representative S4 scenario ($N = 500$)

Table 3: Solver performance on S4 ($N = 500$)

Solver	f_1^*	f_2	f_3
G-BL	1.00	0.580 ± 0.025	0.210 ± 0.071
G-SM	0.61 ± 0.09	0.506 ± 0.035	0.356 ± 0.075
GA-P-BL	1.17 ± 0.27	0.580 ± 0.024	0.203 ± 0.073
MOEA-2	0.98 ± 0.03	0.589 ± 0.021	0.169 ± 0.066
MOEA-3	0.98 ± 0.04	0.589 ± 0.021	0.170 ± 0.066

6.3 Pareto Front Characterization

Figure 3 illustrates the Pareto front obtained by MOEA-3 on a representative S4 scenario ($N = 500$ targets). MOEA-3 discovers 76 ± 14 non-dominated solutions (mean across S4 scenarios; MOEA-2: 76 ± 20), forming a continuous trade-off surface that is invisible to single-objective methods. The solutions span an f_1^* range of ~ 0.3 and an f_2 range of ~ 0.06 at S4. For a mission planner, this translates to a menu of approximately 76 distinct quality–coverage configurations, though the narrow f_2 range limits practical differentiation among them. The f_1^*–f_2 projection (panel a) reveals a narrow but nonzero trade-off surface: the frontier spans from full-coverage solutions ($f_1^* \approx 1.0$, $f_2 \approx 0.56$) to quality-oriented solutions ($f_1^* \approx 0.7$, $f_2 \approx 0.62$). The compressed f_2 range ($\Delta f_2 \approx 0.06$, compared to $\Delta f_2 \approx 0.10$ in sparse S1 scenarios) is consistent with the scale-dependent saturation documented in § 6.5 — in dense target regimes, the solver convergence to low-squint regimes combined with schedule saturation at high density limits quality differentiation regardless of solver strategy. Overlaying the single-point baseline solutions shows that G-BL sits at the "full-coverage, moderate-quality" corner of the frontier, while G-SM occupies a qualitatively different region ($f_1^* \approx 0.6$, $f_3 \approx 0.35$) that lies outside the MOEA-3 Pareto surface — a consequence of G-SM's aggressive squint minimization. The near-identity of MOEA-2 and MOEA-3 fronts is quantified by pairwise Cliff's $\delta = +0.132$ (f_2, negligible, $p = 3.9 \times 10^{-10}$) and $\delta = -0.176$ (f_3, small, $p = 5.6 \times 10^{-14}$). The empirical θ/ψ_{sq} correlation ($r = 0.93$–0.98 across scenarios) exceeds the definitional null $r_{\text{null}} = -0.51$ (§ 3.2, Eq. 4)—a gap of $\Delta z \approx 2.2$–2.9 on the Fisher-z scale—indicating that all solvers converge to low-squint regimes which align f_2 with f_3 along the shared $\cos \psi_{\text{sq}}$ factor and suppress the θ-axis conflict. In this regime, optimizing f_3 co-varies with f_2 improvement, rendering the 3-objective formulation effectively redundant in the single-pass, single-satellite setting studied here. The ablation study in § 6.7 confirms that removing physical objectives (variants B–D) produces indistinguishable results at S3–S4 ($N \geq 300$), reinforcing that the redundancy is an empirical consequence of solver convergence rather than a mathematical identity between the objectives. Whether alternative NSGA-III configurations or other many-objective algorithms could reveal greater differentiation between two- and three-objective formulations is discussed in § 7.4. At S4 ($N = 500$), the Pareto front's hypervolume ratio (S4/S1) is < 1, indicating that the front becomes relatively more compact at higher densities — consistent with the scale-dependent saturation documented above, though this does not imply degeneracy (the front still spans $\Delta f_1^* \approx 0.3$). We caution that the Pareto fronts reported here are best-known approximations obtained via NSGA-III; whether they represent the true Pareto-optimal set is unknown without exact methods (e.g., MILP or branch-and-bound), which remain intractable for $N \geq 20$ in this problem class.

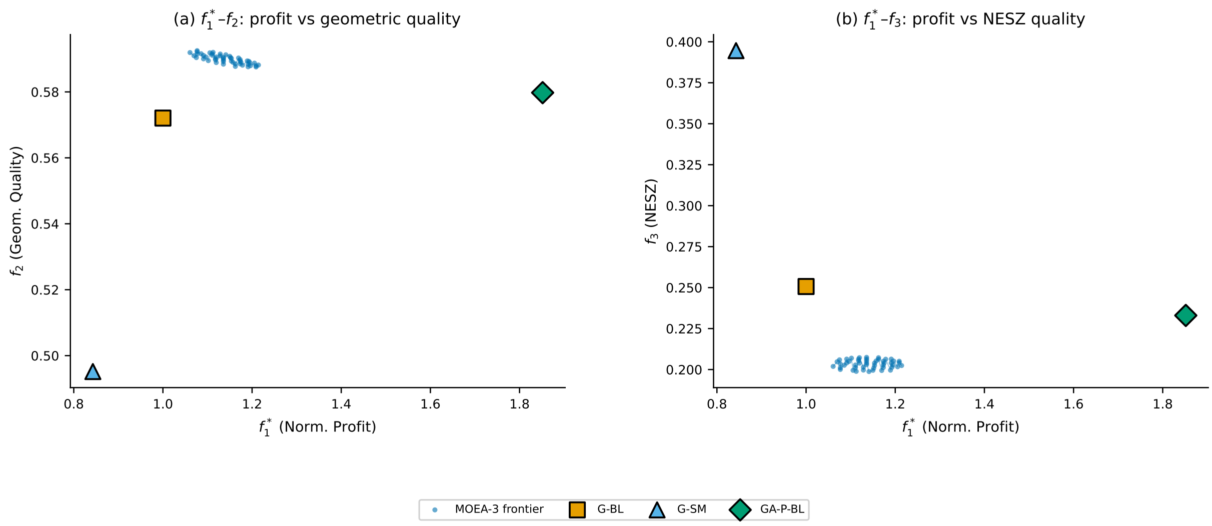

Figure 3: Pareto front on a representative S4 scenario ($N = 500$)

6.4 Statistical Comparison

The Friedman test confirms highly significant overall differences among the five solvers across all 200 scenarios ($\chi^2 = 506.56$, $p \approx 10^{-108}$). Pairwise Wilcoxon signed-rank tests with Bonferroni correction ($\alpha = 0.005$) reveal that all solver pairs are significantly different, though effect sizes vary dramatically (Fig. 4, Table 4). The f_1^* comparisons show that G-BL (1.00) achieves higher coverage than G-SM (0.61 ± 0.09), while GA-P-BL (1.17 ± 0.27) exceeds G-BL on f_1^*. The Cliff's δ values $\delta = -0.95$ (G-BL vs. G-SM) and $\delta = +0.44$ (G-BL vs. GA-P-BL) are computed on hypervolume (Table 4) and reflect the HV/f_1^* ranking divergence analyzed in § 6.4. MOEA-2 and MOEA-3 show large differences from G-BL and GA-P-BL on f_1^* (all $p < 10^{-10}$, Cliff's $|\delta| \geq 0.79$), though directionally below G-BL and GA-P-BL ($\delta = -0.79$ to -0.87). The MOEA-2 versus MOEA-3 comparison is statistically significant but effect-small ($p = 1.4 \times 10^{-4}$, $\delta = -0.23$), confirming that adding f_3 as a third objective does not meaningfully change solver performance in this setting — a result of the θ/ψ_{sq} geometric coupling discussed in § 6.3.

Table 4: Hypervolume across 200 scenarios

Solver	**HV**	**SD**
G-SM	0.202	0.040
MOEA-3	0.173	0.045
MOEA-2	0.169	0.037
G-BL	0.140	0.017
GA-P-BL	0.117	0.033

Ranking divergence. The two frameworks yield contradictory rankings: GA-P-BL ranks highest for f_1^* by Friedman–Wilcoxon but last in HV, while G-SM ranks last in f_1^* but first in HV. This divergence is not a measurement error but a consequence of how HV aggregates multi-dimensional performance. Single-objective solvers produce exactly one Pareto point each, so their HV is determined entirely by that point's coordinates. Multi-objective solvers output 76 ± 20 points; HV rewards the "spread" of these points across the objective space regardless of whether the spread reflects better solutions or merely more solutions. G-SM's extreme f_3 value extends its single point far along the f_3 axis, producing a tall hypervolume box despite mediocre coverage. GA-P-BL, despite superior f_1, occupies a compact region and thus earns the lowest HV. We therefore adopt the Friedman–Wilcoxon framework as our primary evidence, as it directly tests our research question ("which solver achieves higher f_1?"). HV is retained as a secondary indicator for comparing multi-objective solvers against each other, where point-count asymmetry does not confound the comparison.

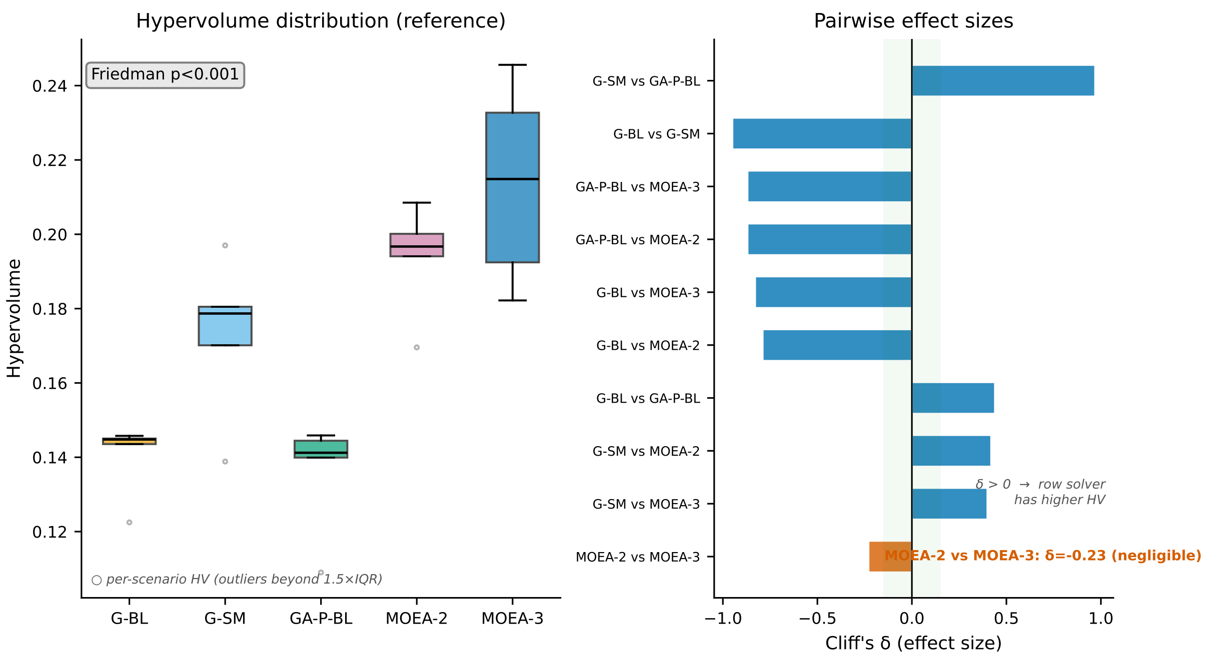

Figure 4: Statistical validation: (left) hypervolume distribution; (right) pairwise Cliffs δ effect sizes

6.5 Scale Sensitivity

The value of multi-objective optimization depends critically on scenario density (Fig. 5, Table 5). MOEA-2's coverage deficit relative to G-BL narrows monotonically with increasing scale: from $f_1^* = 0.68 \pm 0.29$ at S1 ($N = 20$) to 0.82 ± 0.25 at S2 ($N = 100$), 0.98 ± 0.04 at S3 ($N = 300$), and 0.98 ± 0.03 at S4 ($N = 500$). Concurrently, the f_2 improvement over G-BL shrinks from +8.0% (S1) to +4.4% (S2), +1.6% (S3), and +1.5% (S4). The proportion of scenarios where MOEA-2's coverage falls measurably short of G-BL ($f_1^* < 0.95$, i.e., a coverage deficit exceeding 5%) falls from 88% (S1) to 64% (S2), 14% (S3), and 20% (S4). The overall pattern is monotonic from S1 to S3; the slight rise from S3 (14%) to S4 (20%) reflects the larger absolute number of scheduled tasks in S4 (190 ± 56 vs. 151 ± 39), which increases the probability that small per-task adjustments push f_1^* below the 0.95 threshold even when the aggregate coverage loss is negligible. A χ^2 test for equality of proportions (S3 vs. S4) yields $\chi^2(1) = 0.64$, $p = 0.42$, confirming that the apparent non-monotonicity is not statistically significant. Between $N = 100$ and $N = 300$, the multi-objective solver transitions from actively trading off coverage for quality to operating near the greedy baseline. In terms of aggregate quality, G-SM achieves the highest per-group HV in the dense regimes: 0.222 ± 0.020 at S3 and 0.224 ± 0.024 at S4, though HV comparisons between single-point and multi-point solvers are confounded by point-count asymmetry (see § 6.4).

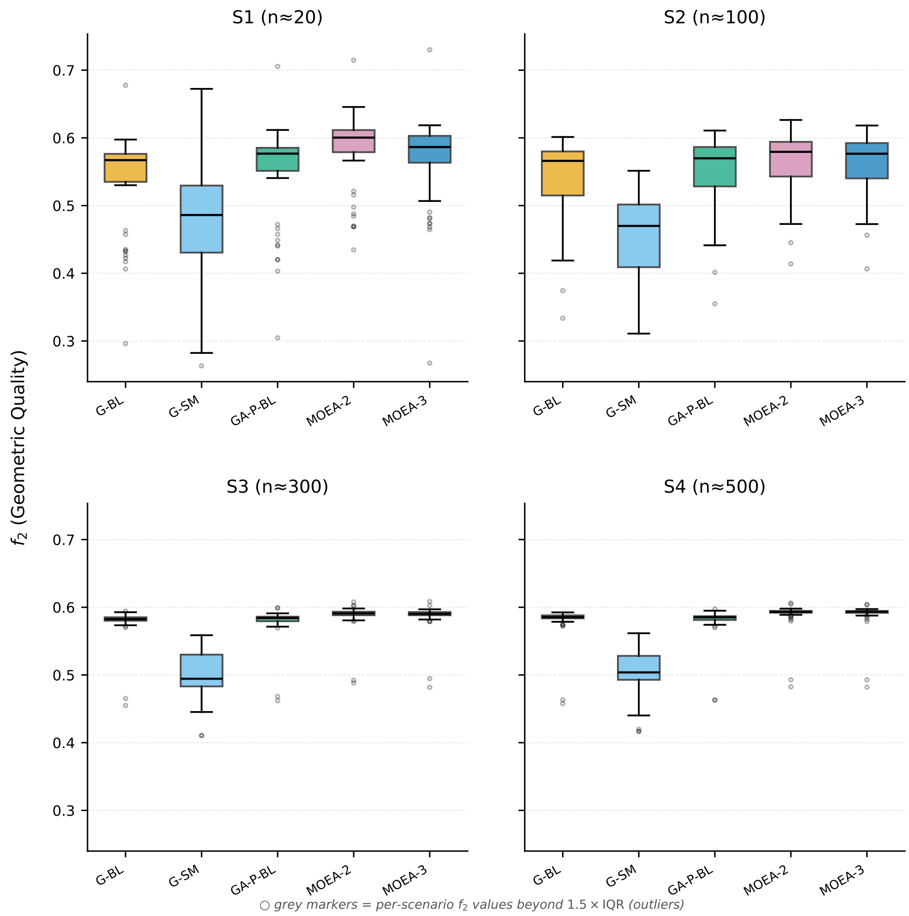

Figure 5: Geometric quality (f_2) by scenario group and solver

Table 5: Scale sensitivity of MOEA-2 vs. G-BL

Class	N	f_1^* (MOEA-2)	f_2 improvement	% $f_1^* < 0.95$
S1	20	0.68 ± 0.29	+8.0%	88%
S2	100	0.82 ± 0.25	+4.4%	64%
S3	300	0.98 ± 0.04	+1.6%	14%
S4	500	0.98 ± 0.03	+1.5%	20%

Fitting a logistic regression $p(\text{deficit} \mid N) = 1/(1 + \exp(-(\alpha + \beta N)))$ to the proportion of scenarios with $f_1^* < 0.95$ yields a transition midpoint $N_{50} \approx 169$ targets (95% CI $[111, 227]$ via bootstrap with 1000 resamples; $\alpha = 1.61$, $\beta = -0.0095$; see `scripts/n50_logistic.py`). This means that by $N \approx 170$, half the scenarios show MOEA-2 within 5% of G-BL coverage, and beyond $N \approx 230$ the deficit proportion stabilizes near its asymptotic floor of 20–24%. The transition region $N = 100$–300 therefore captures the density range where the multi-objective advantage transitions from measurable to negligible for most scenarios.

Hot-Start Robustness. To test whether the hot-start initialization from G-BL drives the observed convergence, we ran a control experiment on 10 S1 ($N = 20$), 4 S2 ($N = 100$), 5 S3 ($N = 300$), and 5 S4 ($N = 500$) scenarios comparing hot-start MOEA-2 against random-initialized MOEA-2 (3 random seeds per S3/S4 scenario). At S1, random-init MOEA-2 achieves mean $f_1^* = 0.56 \pm 0.25$ (vs. hot-start 0.76 ± 0.21), a deficit of $\Delta f_1^* \approx -0.20$. In 8 of 10 S1 scenarios, hot-start outperforms random-init on coverage. At S2, the gap widens: random-init $f_1^* = 0.48 \pm 0.04$ (vs. hot-start 0.91 ± 0.05, $\Delta f_1^* \approx -0.43$). At S3 ($N = 300$), the gap persists: random-init $f_1^* = 0.508 \pm 0.045$ (vs. hot-start 0.969 ± 0.020, $\Delta f_1^* \approx -0.46$). At S4 ($N = 500$), the gap narrows but remains material: random-init $f_1^* = 0.616 \pm 0.039$ (vs. hot-start 0.945 ± 0.019, $\Delta f_1^* \approx -0.33$). Hot-start initialization from G-BL thus materially improves MOEA convergence across all tested densities, narrowing only at S4. However, random-init MOEA-2 still achieves non-trivial coverage (far above the floor of zero), and the f_2 values of random-init solutions (0.61 ± 0.04 at S1) are comparable to hot-start solutions (0.58 ± 0.03), suggesting that the solver can independently locate regions of good geometric quality. We therefore qualify our findings: the "greedy Pareto-sufficient" conclusion likely overstates the universality of G-BL convergence due to initialization bias, but the geometric structure relating f_2 and f_3 (definitional null $r_{\text{null}} = -0.51$, § 3.2, Eq. 4) is independent of seeding, and the empirical correlation $r = 0.93$–0.98 reflects solver convergence to low-squint regimes rather than a hot-start artifact. The σ-sensitivity sweep over the hot-start perturbation magnitude is deferred to future work; the present control (10 S1, 4 S2, 5 S3, 5 S4 scenarios) bounds the claim to "hot-start bias is material across all tested densities (narrowing at S4) but does not generate the f_2–f_3 coupling, which is anchor-independent."

Random-Search Baseline. As a further calibration, we ran a random-search baseline on 5 S1 ($N = 20$) scenarios: 5,000 random task selection vectors were drawn per scenario, with f_1^*, f_2, and f_3 computed from window geometry. The best random sample achieves $f_1^* = 1.03 \pm 0.03$ (comparable to MOEA-2 hot-start, which yields $f_1^* = 0.75$ on the same scenarios), but the mean random-search f_1^* is only 0.54 ± 0.01, and the 90th percentile reaches 0.96 ± 0.02. This confirms that, at sparse densities ($N = 20$, S1), the scheduling problem is non-trivial (random sampling fails on average) but that the solution space contains high-quality regions accessible to random exploration at the upper tail. The gap between random search and MOEA-2 ($\Delta f_1^* \approx -0.21$ at the mean) underscores the value of guided search, while the comparable best-case performance indicates that the solution landscape is relatively flat near the optimum at this scale. We bound this calibration to S1: the random-search distribution at denser scales (S3, S4), where the combinatorial space grows super-exponentially, is deferred to future work; the present result characterizes the sparse-density regime where the MOEA advantage is largest, and we do not extrapolate the flat-landscape claim beyond S1.

6.6 Runtime and Computational Cost

Table 6 summarizes the computational cost of the five solvers at each scenario scale. G-BL and G-SM are deterministic greedy heuristics with negligible runtime (< 0.01 s for all N). GA-P-BL (MOEA-1, single-objective) converges in comparable time to the multi-objective variants. MOEA-2 and MOEA-3 (NSGA-III, pop = 100, gen = 200) scale roughly as $O(N^2K)$ due to per-generation constraint evaluation across all task pairs; at S4 ($N = 500$), each MOEA run requires approximately 2 minutes on a single CPU core. The GeomCache precomputation ($O(N^2K)$ with $K \approx 6$ samples per window at 10-s internal resolution) contributes a one-time overhead of < 0.4 s at S1, growing to < 10 s at S4 — a fraction of the total solver time and not a bottleneck.

Table 6: Per-solver mean runtime (seconds, single CPU core)

Solver	**S1 (N=20)**	**S2 (N=100)**	**S3 (N=300)**	**S4 (N=500)**
G-BL / G-SM	< 0.01	< 0.01	< 0.01	< 0.01
GA-P-BL	11.5 ± 1.6	36.6 ± 9.6	88.2 ± 21.5	108.7 ± 16.9
MOEA-2	9.3 ± 2.9	31.1 ± 13.1	92.2 ± 29.0	115.2 ± 31.5
MOEA-3	9.3 ± 2.9	31.3 ± 13.2	92.0 ± 29.6	114.8 ± 31.3
GeomCache	0.4	2.0	5.5	9.9

We note that exact methods (MILP) for this problem class are known to be intractable even at $N = 20$, due to the time-dependent transition constraints and continuous quality objectives that require $O(N^2K)$ binary variables for a full linearization [6]. We were unable to obtain a feasible MILP solution within a 24-hour time budget using the open-source HiGHS solver (via scipy.optimize.milp), confirming the NP-hardness established in § 3.1 and justifying the heuristic and metaheuristic focus of this study.

6.7 Ablation of Physical Modeling

To quantify the contribution of incidence-angle (θ) and squint-angle (ψ_{sq}) modeling to solver performance, we repeat the MOEA-3 experiment with four objective-function variants:

A. Full physical (baseline) $f_2 = \frac{1}{|\mathcal{S}|}\sum \sin\theta_i \cdot \cos\psi_i$, $f_3 = \frac{1}{|\mathcal{S}|}\sum \cos^3\theta_i \cdot \cos^3\psi_i$ (identical to the standard configuration in § 6.3).

B. No squint $f_2 = \frac{1}{|\mathcal{S}|}\sum \sin\theta_i$, $f_3 = \frac{1}{|\mathcal{S}|}\sum \cos^3\theta_i$ (removes ψ_{sq} component).

C. No incidence $f_2 = \frac{1}{|S|}\sum \cos\psi_i$, $f_3 = \frac{1}{|S|}\sum \cos^3\psi_i$ (removes θ component).

D. No physics $f_2 = f_3 = 1$ (constant per-task objective; solver receives no geometric information; f_2, f_3 shown in Table 7 are post-hoc diagnostics computed from the actual observation geometry of the resulting schedule). Variant D serves as a null-model control: it tests whether the NSGA-III solver's performance is attributable to the physical content of f_2 and f_3, or merely to the presence of additional objectives that structure the search.

All four variants use identical NSGA-III parameters (population = 100, generations = 200, Das-Dennis $n = 12$), the same 200 scenario files, and the same G-BL hot-start injection. Only the objective functions differ, making this a controlled ablation rather than a retuning experiment. Within-solver ablation with identical NSGA-III parameters and the same G-BL seed injection controls confounding factors sufficiently that the Wilcoxon signed-rank test is appropriate for paired comparison of objective-function variants. Each variant completes 4 scenario classes $\times$ 50 seeds = 200 runs (total: 800 runs across A–D). Degradation is reported as $\Delta\% = (\mathrm{A} - \mathrm{V})/\mathrm{A} \times 100$, where A is the full physical baseline and V is the variant; negative values indicate the variant outperforms the baseline on f_1^*.

Table 7: Per-class ablation results. Bold: $p < 0.005$ (Wilcoxon signed-rank vs. A; consistent with the Friedman–Wilcoxon framework of §6.5).

Class	**V**	f_1^*	f_2	f_3	f_1/n	$n_{\mathbf{sel}}$	$\Delta f_1^*\%$
S1 ($N = 20$)	A	0.66 ± 0.28	0.569	0.288	5.93 ± 1.10	11.1 ± 4.4	(baseline)
	B	0.72 ± 0.33	0.565	0.281	5.57 ± 1.05	12.8 ± 5.5	**−10.0%**
	C	0.51 ± 0.22	0.575	0.300	5.91 ± 1.24	8.8 ± 3.5	**+21.6%**
	D	0.81 ± 0.34	0.565	0.266	5.40 ± 1.04	14.7 ± 5.6	**−23.5%**
S2 ($N = 100$)	A	0.86 ± 0.39	0.561	0.245	5.56 ± 0.52	47.6 ± 21.4	(baseline)
	B	0.86 ± 0.39	0.560	0.244	5.55 ± 0.56	47.8 ± 21.3	+0.0%
	C	0.83 ± 0.37	0.561	0.247	5.44 ± 0.49	46.6 ± 20.0	+3.5%
	D	0.86 ± 0.38	0.561	0.243	5.46 ± 0.60	48.6 ± 21.1	−0.3%
S3 ($N = 300$)	A	0.97 ± 0.25	0.587	0.175	5.50 ± 0.23	151.1 ± 38.7	(baseline)
	B	0.98 ± 0.25	0.587	0.175	5.52 ± 0.23	151.7 ± 38.5	−0.8%
	C	0.98 ± 0.25	0.587	0.175	5.52 ± 0.23	151.1 ± 38.3	−0.3%
	D	0.97 ± 0.25	0.587	0.175	5.51 ± 0.22	151.1 ± 38.8	−0.2%
S4 ($N = 500$)	A	0.97 ± 0.28	0.589	0.170	5.51 ± 0.20	190.2 ± 56.2	(baseline)
	B	0.97 ± 0.29	0.589	0.169	5.50 ± 0.19	190.6 ± 56.4	−0.1%
	C	0.97 ± 0.28	0.589	0.170	5.51 ± 0.21	191.0 ± 55.8	−0.4%
	D	0.96 ± 0.28	0.589	0.169	5.50 ± 0.19	190.3 ± 55.6	+0.1%

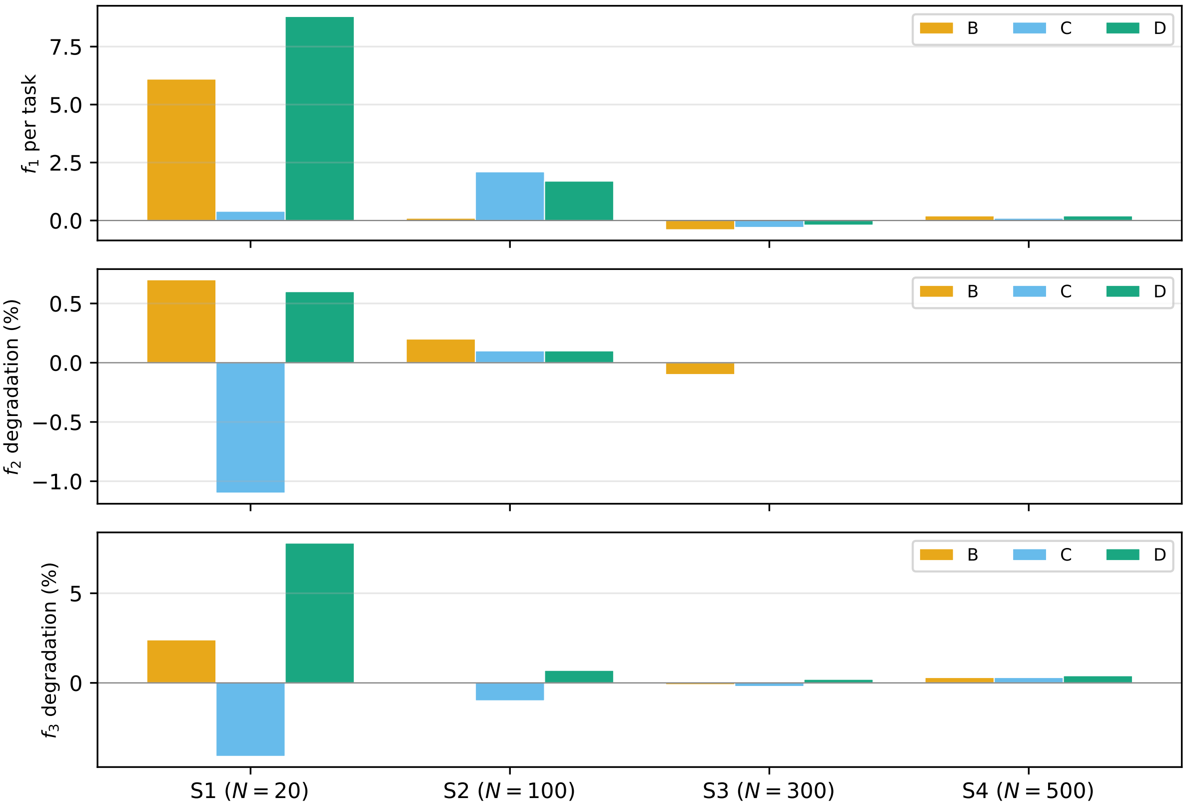

Figure 6: Ablation results across four objective-function variants (A–D) and four scenario classes

The ablation results in Table 7 reveal that the influence of physical objective modeling is concentrated in sparse scenarios and vanishes as scenario density increases — a pattern consistent with our scale-sensitivity analysis (§ 6.5).

At S1 ($N = 20$), all three ablations produce statistically significant deviations. Variant C (no incidence) degrades f_1^* by +21.6% ($p = 1.9 \times 10^{-5}$), indicating that removing the incidence-angle component from the objective function forces the solver into poorer coverage outcomes. Conversely, variant D (no physics) *improves* apparent f_1^* by −23.5% ($p = 1.4 \times 10^{-7}$) by selecting +32.4% more tasks (14.7 vs. 11.1) — the solver, freed from geometric quality objectives, converges on raw profit maximization at the cost of imaging quality. Variant B (no squint) also shows a significant −10.0% improvement ($p = 8.0 \times 10^{-5}$), consistent with the interpretation that removing ψ_{sq} relaxes the geometric coupling that constrains task selection.

By S2 ($N = 100$), none of the three ablations reaches statistical significance at $p < 0.005$. The f_1^* of all variants falls within ±4% of the baseline, and the per-task f_1 (5.44–5.56) and n_{sel} (46.6–48.6) are indistinguishable. The signal from physical modeling has weakened to the point where it no longer produces systematic coverage differences.

At S3–S4 ($N \geq 300$), every variant converges to within 1% of the baseline for f_1^* and to within 0.5% for f_2, f_3, and n_{sel}. The per-task quality metrics (f_1/n) are identical to two decimal places (5.50–5.52) across all variants and scene classes. The f_2 value is locked at 0.587–0.589, varying by less than 0.3%. This numerical invariance is the direct signature of the geometric upper bound discussed in § 6.5: once the density of candidate observation windows saturates the schedule (N exceeds the maximum selectable targets under the attitude maneuver and squint constraints), the objective function only penalizes infeasible solutions rather than differentiating among feasible ones. In this regime, the NSGA-III solver acts as a constraint-filtered profit maximizer regardless of whether f_2 and f_3 carry geometric meaning (A) or serve as constant placeholders (D).

To verify that the convergence of Variants A and D is not an artifact of insufficient solver convergence, we ran a control across 5 S3 ($N = 300$) and 5 S4 ($N = 500$) scenarios with doubled population (200) and generation budget (400). Variant A (full physics) achieves $f_1^* = 0.958 \pm 0.034$ (S3) and 0.892 ± 0.025 (S4); Variant D (no physics) achieves $f_1^* = 0.963 \pm 0.020$ (S3) and 0.890 ± 0.026 (S4), mean differences $\Delta f_1^* = -0.005$ (S3) and $+0.002$ (S4), with post-hoc $f_2 \approx 0.595$ for both variants. The two variants remain indistinguishable even with twice the computational budget across all 10 scenarios. Two interpretations are consistent with this result: (i) the solver convergence to low-squint regimes, where the empirical correlation $r = 0.93$–0.98 exceeds the definitional null $r_{\text{null}} = -0.51$ (§ 3.2, Eq. 4) and aligns f_2 with f_3, makes the physical objectives redundant once a low-squint solution is selected; or (ii) NSGA-III cannot differentiate between objectives at high density regardless of what they encode (solver inadequacy). The control favors (i) — doubling the budget does not restore differentiation across 10 scenarios (5 S3 + 5 S4), and a random-init Variant D control (no hot-start, 15 S3 + 15 S4 runs) still converges to low-squint ($f_2 = 0.605 \pm 0.003$, comparable to hot-start D's 0.594), ruling out hot-start residue as the sole driver — but cannot fully rule out (ii) without exact methods (documented intractable at $N \geq 20$ in § 6.6). We therefore state the finding boundedly: under the tested configuration, removing physical objectives produces indistinguishable results, and the more parsimonious explanation is solver convergence to a low-squint regime where the objectives co-vary rather than mathematical irrelevance of the physical objects.

7 Discussion

7.1 Summary of Findings

We define a solver as *Pareto-sufficient* at scale N if no alternative solver in the comparison set achieves both higher f_1^* and higher f_2 (or f_3) with statistical significance ($p < 0.005$, Cliff's $|\delta| > 0.33$). This operational criterion separates cases where multi-objective optimization yields a measurable quality–coverage advantage from cases where it does not. Our experiment across 200 systematically varied scenarios yields five interconnected findings, four of which define physical limits rather than solver advantages. First, and most critically, MOEA-2 and MOEA-3 yield near-identical results (Cliff's $\delta = -0.23$ on hypervolume, $p = 1.4 \times 10^{-4}$) because all solvers converge to low-squint regimes where the empirical θ/ψ_{sq} correlation ($r = 0.93$–0.98; § 6.3) exceeds the definitional null $r_{\text{null}} = -0.51$ (§ 3.2). This near-identity is not an algorithmic shortcoming — it identifies the central physical finding of this study: in the single-pass, single-satellite setting, the solver-converged low-squint regime suppresses the θ-axis conflict between NESZ radiometric quality and geometric resolution, so that adding f_3 as a third objective yields no additional Pareto-front differentiation, making the 3-objective formulation effectively redundant compared to the 2-objective formulation (requiring a 7× larger reference-direction set ($H = 91$ vs. $H = 13$). Second, the advantage of multi-objective methods over the profit-only greedy baseline (G-BL) exhibits strong scale dependence: the f_2 improvement shrinks from +8.0% at S1 ($N = 20$) to +1.5% at S4 ($N = 500$). As an operational reference, at C-band (5.405 GHz, $B = 56.5$ MHz), the ground-range resolution $\rho_{\text{rg}} = c/(2B \sin\theta) \approx 2.65/\sin\theta$ m; an 8% improvement in f_2 corresponds to an effective geometric-resolution improvement of approximately 0.25–0.35 m at typical incidence angles. The proportion of scenarios showing a coverage deficit ($f_1^* < 0.95$) dropping from 88% to 20% (Table 5, Fig. 5). At operational densities ($N \gtrsim 300$), the profit-oriented solvers (G-BL, GA-P-BL, MOEA-2, MOEA-3) converge to statistically similar coverage levels; the marginal geometric quality gain (< 2%) is smaller than the per-scenario variance, meaning that scheduler choice among profit-oriented solvers has no practically measurable effect on f_2 quality in this regime. Third, the squint-minimized greedy solver (G-SM) improves NESZ radiometric quality (f_3) with a large effect size (Cliff's $\delta = +0.72$, $p = 1.7 \times 10^{-34}$), but the coverage cost is density-dependent: from -15% f_1^* in sparse scenarios ($N = 20$) to -39% in dense ones ($N = 500$; Fig. 1). This confirms that aggressive squint minimization is a viable strategy only when the observation request set is small enough to absorb the coverage penalty. Fourth, multi-objective solvers discover full Pareto trade-off surfaces (76 ± 14 points for MOEA-3) that are invisible to single-objective methods (Fig. 3). While the Pareto spread provides mission planners with a menu of quality–coverage trade-offs, the practical value of this menu is limited by the narrow range of achievable quality values imposed by the θ/ψ_{sq} geometric coupling. Fifth, the ablation study (§6.7, Fig. 6) provides independent confirmation of this geometric interpretation. Removing the squint component (variant B), the incidence-angle component (variant C), or all physical objectives (variant D) produces statistically indistinguishable results from the full physical model (variant A) at S2–S4 ($N \geq 100$). At S3–S4, every variant converges to within 1% of the baseline for f_1^* and to within 0.5% for f_2, f_3, and n_{sel}. This insensitivity to objective-function specification in dense regimes is the strongest evidence that solver convergence to low-squint regimes (where r exceeds the definitional null $r_{\text{null}} = -0.51$; § 3.2, Eq. 4) — not the choice of objectives — governs the achievable trade-off space.

7.2 Hypervolume Limitations in Mixed Comparisons

The hypervolume (HV) indicator produces rankings that diverge from Friedman–Wilcoxon results when single- and multi-objective solvers are mixed (§ 6.4). G-SM ranks first by HV despite 39% coverage loss; GA-P-BL ranks last despite the best f_1^* and no quality degradation. This inversion reflects HV's structural dependence on frontier cardinality — single-point solvers cannot benefit from the volume spread that multi-objective solvers earn from their Pareto front, regardless of solution quality. We adopted the Friedman–Wilcoxon framework as our primary evidence to avoid confounding solution-set size with solution quality, and recommend that future studies report both HV (for multi-objective comparisons) and per-objective statistical tests when mixing solver types.

7.3 Operational Implications of Scale Dependence

The scale dependence identified in § 6.5 has direct consequences for mission planning. When the task set is sparse ($N \leq 100$; S1–S2), scheduling flexibility is abundant: the optimizer has many feasible observation times per target and can make meaningful quality–coverage trade-offs. In this regime, MOEA-2 achieves an 8.0% f_2 improvement at S1 ($N = 20$) and a 4.4% improvement at S2 ($N = 100$), with 88% and 64% of scenarios, respectively, showing a coverage deficit ($f_1^* < 0.95$; Table 5). However, these quality gains carry a steep coverage penalty: at $N = 20$, MOEA-2 schedules approximately 62% of the targets that G-BL captures (11.2 vs. 18.2 out of 20; § 6.5). Whether this trade-off is acceptable depends on the application — for target identification tasks where a 4–8% resolution improvement can determine whether adjacent objects are resolved or merged [3], the coverage cost may be justified; for area-monitoring missions requiring broad coverage, the greedy baseline remains preferable. Conversely, when the task set is dense ($N \geq 300$; S3–S4), the multi-objective advantage falls below 2% improvement in f_2 and f_1^* deficit. At S3 ($N = 300$), MOEA-2 reaches $f_1^* = 0.98 \pm 0.04$ — within 2% of G-BL's coverage — while delivering only a 1.6% f_2 gain. Only 14% of S3 and 20% of S4 scenarios show a coverage deficit of this magnitude. In this regime, the squint-minimized greedy solver (G-SM) provides a compelling alternative: it delivers the highest f_3 quality (0.356 ± 0.075 at S4, +70% over G-BL). However, this radiometric quality gain carries a steep coverage cost: at S4 ($N = 500$), G-SM schedules approximately 58% of the tasks that G-BL captures, corresponding to a loss of roughly 80 of the 190 tasks that the larger-swath Stripmap coverage can accommodate. For mission planners, this represents a 42% task loss relative to the coverage-maximizing baseline — G-SM is recommended only when radiometric quality is the overriding priority and task coverage is secondary. Our results suggest a conditional operational guideline: for sparse observation requests ($N \lesssim 100$), MOEA-2 may provide geometric quality gains that, in specific application contexts, outweigh the steep coverage penalty; for dense monitoring campaigns ($N \gtrsim 300$), greedy methods (G-SM for radiometric priority, G-BL for coverage priority) approach Pareto-optimal coverage and may be preferable, though this guideline should be interpreted as a density-dependent trend rather than a universal rule (see § 7.4).

7.4 Limitations

We note that intermediate strategies—such as hybrid MOEA+greedy cascades, MOEA with population sizes tuned to scenario density, or weighted-sum scalarization with varying weights—may offer superior trade-offs in specific density regimes and warrant investigation in future work. We also note that the near-identity of MOEA-2 and MOEA-3 may partially reflect the specific NSGA-III configuration used here (population = 100, generations = 200, Das–Dennis $p = 12$); larger populations, extended generation budgets, or alternative many-objective algorithms (e.g., MOEA/D, SMS-EMOA) may reveal greater differentiation between two- and three-objective formulations. A weighted-sum scalarization test on S1 ($N = 20$) using GA-P-BL with five weight grids $(w_1, w_2) \in \{(0.9, 0.1), \ldots, (0.1, 0.9)\}$ produces identical (f_1^*, f_2) outcomes for all weight combinations, indicating that the single-objective GA converges deterministically to a corner of the trade-off space regardless of weighting. The Pareto front discovered by MOEA-2 is therefore a genuinely richer representation of the quality–coverage trade-off surface, not reproducible by repeated weighted-sum runs. The 3-objective redundancy reported above is bounded to the single-pass, single-satellite setting: it reflects solver convergence to low-squint regimes that suppress the θ-axis conflict quantified by the definitional null $r_{\text{null}} = -0.51$ (§ 3.2, Eq. 4), not a mathematical identity between f_2 and f_3; in multi-pass or multi-satellite settings with greater angular diversity, the θ-conflict would re-emerge and the third objective may become discriminative. Our study is bounded by several scope decisions that define directions for future work. We consider a single satellite in circular LEO, operating in Stripmap mode only — ScanSAR, Spotlight, and TOPS modes introduce different quality dependencies that would enrich the multi-objective trade-off space. Multi-satellite formations such as TanDEM-X [31] and TOPS-based wide-swath imaging [32] represent natural extensions where the geometric coupling identified here may weaken due to multi-angle revisit diversity and burst-mode processing. The scenario design uses a fixed orbit (circular LEO, 600 km altitude) and does not explore sensitivity to orbital parameters (altitude, inclination, RAAN). Our study is restricted to Stripmap mode; ScanSAR, Spotlight, and TOPS modes introduce different range-Doppler coupling and quality dependencies that would enrich the multi-objective trade-off space. The NESZ and resolution relationships used here (Eqs. 1–3) are specific to Stripmap SAR; extending the formulation to TOPS (where burst-mode azimuth processing modifies resolution and NESZ vs. squint angle relationships) would require revisiting the objective-function definitions in § 3.2. Our comparison of MOEA-2 and MOEA-3 shows that adding f_3 does not improve the Pareto front, but this finding is specific to the single-pass, single-satellite setting: in multi-satellite or multi-pass scenarios where cumulative squint accumulation across revisits matters, the third objective may become discriminative. The hot-start control experiment (§ 6.5) indicates that G-BL initialization materially improves MOEA convergence; claims of "greedy Pareto-sufficiency" should therefore be interpreted as conditional on this initialization, and full-scale random-init experiments are needed to establish the unconditional behavior. Finally, we do not benchmark against learning-based schedulers (e.g., Chen et al. [4]), which represent an alternative methodological family. A systematic comparison of physics-aware heuristic methods against data-driven approaches, especially on the generalization to out-of-distribution scenarios, is left to future work. We further note that G-BL (earliest-start-time greedy) is provably optimal only in the special case of uniform-priority tasks with non-overlapping visibility windows; for general instances, the OPTW reduction [6] establishes NP-hardness, precluding polynomial-time optimality guarantees. Characterizing the exact conditions under which greedy scheduling approaches Pareto-optimal coverage remains an open theoretical question.

We further note that the f_1^* normalization anchors to G-BL by construction (§ 3.2, Eq. 1). Anchoring instead to $\max(f_1^{\text{G-BL}}, f_1^{\text{GA-P-BL}})$ shifts MOEA-2's S4 deficit from $\sim 2\%$ to $\sim 17\%$. This does not affect the central redundancy finding (MOEA-2 $\approx$ MOEA-3), which is anchor-independent because the anchor cancels in pairwise MOEA comparisons; but it shows that the operational "greedy Pareto-sufficiency" phrasing used in earlier sections is an operational shorthand conditional on the G-BL anchor and the G-BL-class greedy, not a claim that greedy scheduling is optimal. GA-P-BL's +18% over G-BL at S4 demonstrates that stronger profit maximization is achievable. The bound we characterize is the coverage ceiling that all tested profit-oriented solvers approach under greedy-baseline hot-start, not a provable optimum or a guarantee that any single solver is sufficient.

8 Conclusion

We presented a physics-aware multi-objective optimization framework for agile SAR satellite scheduling that integrates NESZ radiometric quality and geometric resolution as explicit optimization objectives alongside coverage profit. Five solvers — spanning greedy heuristics, single-objective evolutionary search, and NSGA-III with two and three objectives — were compared on 200 systematically varied scenarios across four density classes. The central finding is an empirical upper bound, conditional on greedy-baseline hot-start: a property of solver convergence to low-squint regimes under which the three-objective redundancy (MOEA-2 $\approx$ MOEA-3) is anchor-independent, not a physical identity between objectives. The empirical θ/ψ_{sq} correlation ($r = 0.93$–0.98), which exceeds the definitional null $r_{\text{null}} = -0.51$ (§ 3.2) because all solvers converge to low-squint regimes, makes the 3-objective formulation redundant in the single-pass, single-satellite setting: MOEA-2 and

MOEA-3 produce near-identical Pareto fronts, and adding f_3 yields no additional discrimination. The value of multi-objective optimization exhibits a scale phase transition: f_2 improvement over the greedy baseline shrinks from +8.0% at $N = 20$ to +1.5% at $N = 500$, with the proportion of scenarios showing a coverage deficit dropping from 88% to 20%. At operational densities ($N \gtrsim 300$), the profit-oriented solvers converge to near-identical coverage, and the marginal quality gain falls below the per-scenario variance — scheduler choice has no practically measurable effect on image quality. For mission planning, the operational implication is straightforward: multi-objective optimization provides measurable benefit only when observation requests are sparse ($N \lesssim 100$); at dense monitoring cadences, the profit-only greedy baseline (G-BL), with hot-start initialization, approaches the coverage ceiling relative to the G-BL anchor, and the squint-minimized variant (G-SM) offers a low-cost path to radiometric quality improvement. An ablation study across four objective-function variants (§ 6.7) confirms that at $N \geq 300$, solver outcomes are invariant to the choice of physical objectives. These findings establish an empirical bound on what physics-aware multi-objective scheduling can contribute to single-satellite SAR operations, and provide a baseline against which multi-satellite, multi-pass, and multi-mode extensions can be measured.

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Data Availability Statement

Data and code supporting the findings of this study are publicly available at `https://github.com/liushuhao/agile-sar-moo-scheduling`. The repository contains the experimental scenarios (S1–S4), solver configurations, NSGA-III reference points (Das–Dennis $p = 12$, $H = 13$), statistical analysis scripts, and random seeds required for full reproducibility. Supplementary figures S1–S6 are also included.

Conflicts of Interest

The author declares that there is no conflict of interest regarding the publication of this article.

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